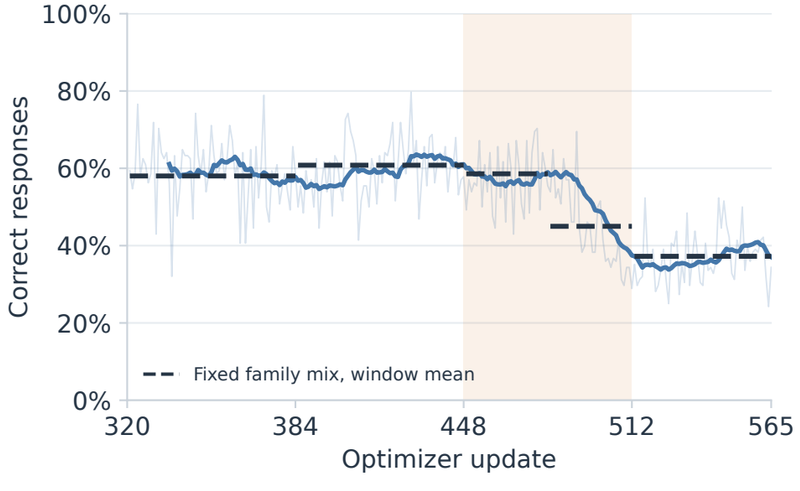
The policy becomes less stable before accuracy collapses

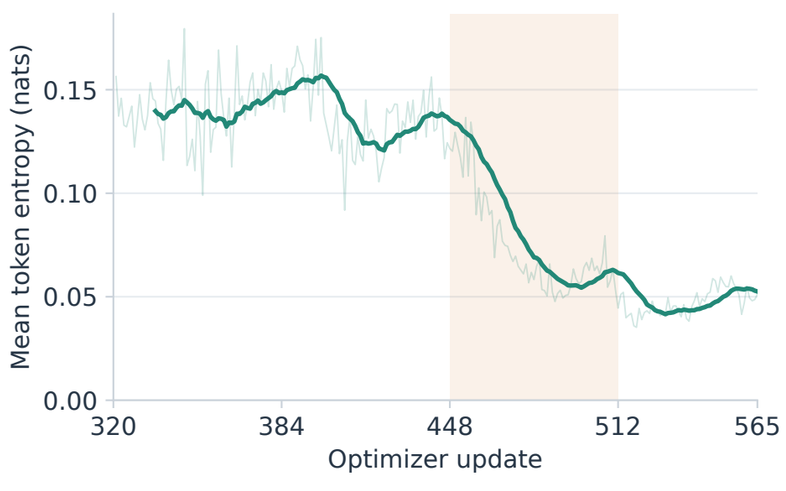
245 updates on 31,360 training responses • faint: each update • dark: trailing 16-update mean

Shading marks steps 448–512, the interval containing the sharp validation drop; it is not an estimated change point.

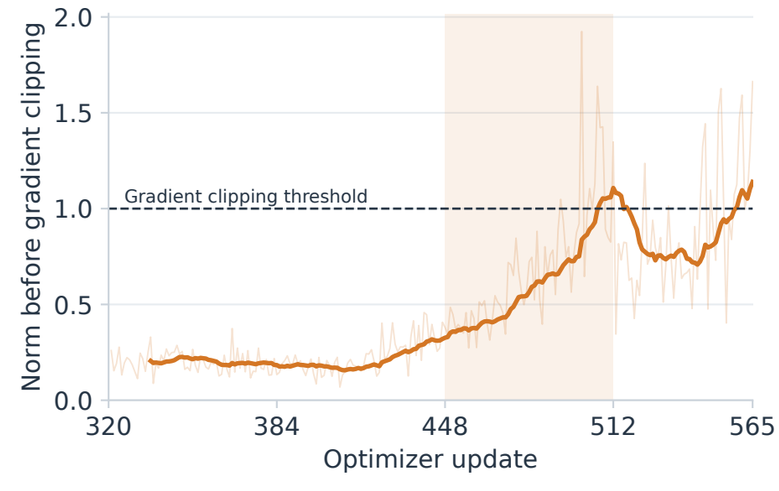
A Training accuracy also falls



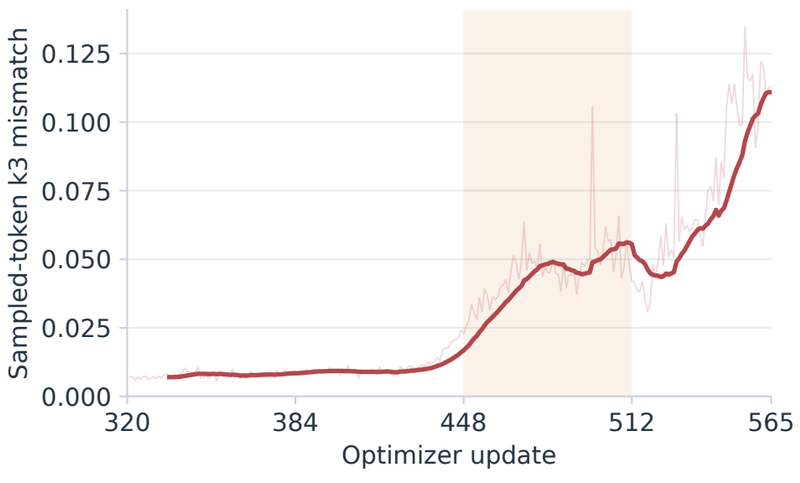
B Policy entropy falls



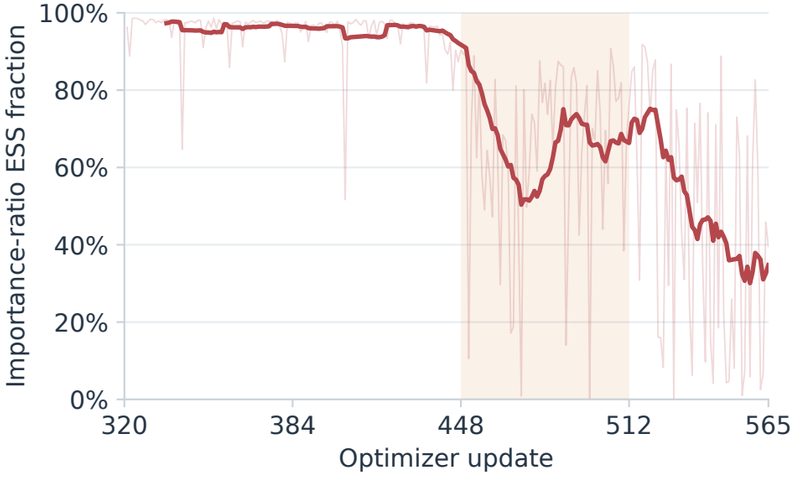
C Gradient norms rise



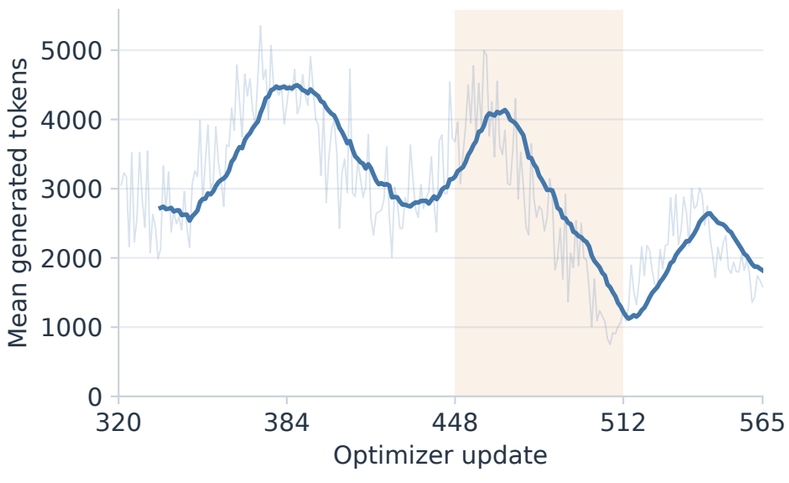
D Train / rollout mismatch grows



E Probability ratios concentrate



F Training responses shorten



PPO clipping activated on 0% of tokens in every audited update; one optimizer pass per batch. Reference KL, entropy loss and rollout correction were disabled.
Mismatch is between Megatron training and vLLM rollout probabilities, not KL to the original model. ESS is a ratio diagnostic, not independent task count.
Fixed-family weights use steps 321–448, retaining 99.9% of reference responses. Within-family difficulty is uncontrolled. These are observational diagnostics.